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Decision Making under Uncertainty in Financial Markets

Improving Decisions with Stochastic Optimization

Jonas Ekblom



Department of Management and Engineering
Division of Production Economics
Linköping University, SE-581 83 Linköping, Sweden

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Abstract

This thesis addresses the topic of decision making under uncertainty, with particular focus on financial markets. The aim of this research is to support improved decisions in practice, and related to this, to advance our understanding of financial markets. Stochastic optimization provides the tools to determine optimal decisions in uncertain environments, and the optimality conditions of these models produce insights into how financial markets work. To be more concrete, a great deal of financial theory is based on optimality conditions derived from stochastic optimization models. Therefore, an important part of the development of financial theory is to study stochastic optimization models that step-by-step better capture the essence of reality. This is the motivation behind the focus of this thesis, which is to study methods that in relation to prevailing models that underlie financial theory allow additional real-world complexities to be properly modeled.

The overall purpose of this thesis is to develop and evaluate stochastic optimization models that support improved decisions under uncertainty on financial markets. The research into stochastic optimization in financial literature has traditionally focused on problem formulations that allow closed-form or ‘exact’ numerical solutions; typically through the application of dynamic programming or optimal control. The focus in this thesis is on two other optimization methods, namely stochastic programming and approximate dynamic programming, which open up opportunities to study new classes of financial problems. More specifically, these optimization methods allow additional and important aspects of many real-world problems to be captured.

This thesis contributes with several insights that are relevant for both financial and stochastic optimization literature. First, we show that the modeling of several real-world aspects traditionally not considered in the literature are important components in a model which supports corporate hedging decisions. Specifically, we document the importance of modeling term premia, a rich asset universe and transaction costs. Secondly, we provide two methodological contributions to the stochastic programming literature by: (i) highlighting the challenges of realizing improved decisions through more stages in stochastic programming models; and (ii) developing an importance sampling method that

can be used to produce high solution quality with few scenarios. Finally, we design an approximate dynamic programming model that gives close to optimal solutions to the classic, and thus far unsolved, portfolio choice problem with constant relative risk aversion preferences and transaction costs, given many risky assets and a large number of time periods.

Sammanfattning

I denna avhandling studeras beslutsfattande under osäkerhet med fokus på finansiella marknader. Forskningens syfte är att understödja bättre beslut i praktiken och därigenom att bidra till bättre förståelse av finansiella marknader. Området stokastisk optimering ger verktygen för att bestämma optimala beslut under osäkerhet och optimalitetsvillkoren från dessa modeller ger centrala insikter om hur finansiella marknader fungerar. Mer konkret bygger mycket av finansiell teori på optimalitetsvillkor från stokastiska optimeringsmodeller. En viktig del i vidareutvecklandet av finansiell teori är därför att studera stokastiska optimeringsmodeller som stegvis bättre fångar centrala egenskaper från verkligheten. Detta motiverar avhandlingens fokus, vilket är att studera metoder som i relation till rådande modeller som ligger till grund för finansiell teori erbjuder ytterligare aspekter från verkligheten att modelleras.

Det övergripande syftet med denna avhandling är att utveckla och utvärdera stokastiska optimeringsmodeller som understödjer bättre beslut under osäkerhet på finansiella marknader. Forskningen om stokastisk optimering i finansiell litteratur har traditionellt varit fokuserad på problemformuleringar som möjliggör analytiska eller 'exakta' numeriska lösningar; typiskt genom tillämpning av dynamisk programmering eller kontrollteori (optimal control). Denna avhandling fokuserar på två andra optimeringsmetoder, nämligen stokastisk programmering och approximativ dynamisk programmering, vilka öppnar upp möjligheter att studera nya klasser av finansiella problem. Mer specifikt möjliggör dessa optimeringsmetoder att fler viktiga aspekter i många finansiella problem kan tas hänsyn till.

Avhandlingen bidrar med flera insikter av relevans för litteraturen inom både finans och stokastisk optimering. Först visar vi att modellering av flera centrala aspekter från verkligheten som traditionellt inte inkluderats i litteraturen utgör viktiga komponenter i en modell som understödjer företags riskhanteringsbeslut. Mer specifikt påvisar vi betydelsen av att modellera räntepremier, ett rikt univers av hedginginstrument och transaktionskostnader. Därefter lämnar avhandlingen två metodmässiga bidrag till litteraturen inom stokastisk programmering i form av att vi: (i) belyser utmaningarna med att realisera bättre beslut genom att addera tidssteg i stokastiska programmeringsmodeller; och (ii)

utvecklar en metod för scenariegenerering baserad på byte av sannolikhetsmått (importance sampling). Vi visar att den föreslagna scenariegenereringsmetoden kan användas för att erhålla beslut av hög kvalité givet endast ett fåtal scenarier. Slutligen designar vi en approximativ dynamisk programmeringsmodell som bestämmer näroptimala lösningar till det klassiska, men dittills olösta, portföljvalsproblemet med CRRA preferenser och transaktionskostnader, givet många tillgångar och ett stort antal tidsperioder.

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Finally, my wife Sofia and my daughter Wilma, for all their love and for being the best at making me happy. I love you more than you can imagine!

Linköping, September 17, 2018

Jonas Eklom

List of Papers

The following papers are appended and will be referred to by their numerals.

- I.** Blomvall, J., Ekblom, J. (2018). Corporate Hedging: An Answer to the “How” Question. *Annals of Operations Research* 266 (1-2):35–69.
- II.** Birge, J.R., Blomvall, J. Ekblom, J. (2018). On the Value and Cost of More Stages in Stochastic Programming. *Working paper*.
- III.** Ekblom, J., Blomvall, J. (2018). Stochastic Optimization with Importance Sampling: Using an Analytical Approximation of the Zero-Variance Distribution. *This paper is under revision*.
- IV.** Blomvall, J. Ekblom, J., Birge, J.R. (2018). An Approximate Dynamic Programming Model for Dynamic Portfolio Choice with Transaction Costs. *Working paper*.

Author Statements

Paper I

The first author proposed the main research idea and the major design of the stochastic programming framework. The term structure estimation method utilized in this study is also from the first author. The second author contributed with a number of methodological ideas and had the main responsibility for deriving the necessary mathematics, realizing the computer implementations and writing the paper.

Paper II

The first author provided feedback during the research phase and on the written manuscript. The second author identified the research gap filled by the paper and provided feedback both during the research phase and on the written manuscript. The third author contributed with the research design, and he also derived the underlying mathematics, implemented the models, and wrote the paper.

Paper III

The first author developed the main research idea, realized the computer implementations and wrote the paper. The second author provided valuable perspectives on the structuring of results, on the scenario generation for the multi-stage stochastic programming model, and with feedback on the written manuscript.

Paper IV

The first author contributed to the design of the ADP model and provided feedback on the written manuscript. The second author contributed with methodological ideas, derived the underlying mathematics, implemented the model and wrote the paper. The third author contributed with feedback on the written manuscript.

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Introduction

I would like to start by asking you to

Think about two important decisions you made over the last year.

Maybe your decisions were about: which car, apartment or house to buy; how to invest your pension capital; or which school and study program to choose? A common theme of these decisions, and most likely yours, is that they are made under uncertainty. Even an everyday decision of whether to take a raincoat is affected by uncertainty. Maybe you do not explicitly think about these as decisions under uncertainty, but they are, and so are many more of all the types of decisions you make every day. Considering all these decisions made under uncertainty, I would next like to ask you:

How do you make decisions under uncertainty?

May I guess that you in some way actually have considered the uncertainty? For example in terms of the likelihood of a Friday car, a jerry-built building or of not getting a good job after graduation. Maybe your decision process was guided by the perception of good and bad outcomes and the associated likelihood of these being realized? If we consider the raincoat decision as an example and assume that taking the raincoat is inconvenient (a cost) for you, but that this is more than compensated in case nature gives us rain. Your decision does then (naturally) depend on the relationship between the probability of rain and the inconvenience of carrying the coat; but how can we judge whether a decision is good or bad? This leads us to my third question:

Did you make good decisions?

What is your argument for whether the decisions were good or bad? Is it because you decided not to take your rain coat and it did not rain or because

the newly bought car has not yet broken down? To challenge your thoughts, consider a drunk person deciding to drive home after a party, and assume that the trip is accident-free. In line with the reasoning above, does this imply that it was a good decision? Obviously not, and the reason is that when we study a large number of trips with drunk drivers, the relative number of accidents is significantly higher than for those with sober drivers. The point here is that a single realization of uncertainty does not provide a good basis for judging the quality of a decision made under uncertainty. The reason is that a single realization is too strongly affected by randomness. Further, our perception of uncertainty is most often incorrect, which also makes this an unsuitable basis for evaluating the quality of a decision. Proper evaluation hinges instead on the possibility of observing a large number of realizations of uncertainty, from which we can observe the true range of consequences from different decisions.

The discussion so far summarizes what this thesis is about, namely the study of decision making under uncertainty based on models of possible outcomes and associated probabilities, from which the goal is to obtain good decisions. The appended studies focus on decision problems on financial markets, where uncertainty plays a key role. We study for example the management of risk for a global company with exposure to foreign currencies and interest rates in different markets. Further, we investigate the problem of asset management, i.e., how to invest money in a ‘good’ way. Our goal is to obtain decisions that are good in practice, and to this end we investigate different aspects of how to model and formulate these problems. As we now move on, I ask you to keep the above discussion in your mind. While the road towards making good decisions in the problems that we study is both challenging and demanding, the goal should appeal to all of us—improving decisions.

1.1 Background

As the introduction indicates, this thesis studies decision making under uncertainty in financial markets. The main scientific tools for this purpose are from the field of stochastic optimization. While for the decisions of an individual discussed above ‘good enough’ decisions are often obtained without the use of mathematical models, many problems on financial markets are too complex in their nature to be properly analyzed without mathematical support. The motivation for developing mathematical models to support decision making is, however, not its complexity alone, but the combination of complexity with a large potential impact on society of these decisions. There are many examples of bad decisions in complex problems on financial markets that have led to significant losses for society, not least the credit crisis that escalated with the collapse of Lehman Brothers in 2008.

Stochastic optimization models produce decisions that are (ideally) optimal given a necessarily simplified model of reality. The best decisions given by

the model are, however, by no means guaranteed to be good in practice. While we cannot hope to model reality exactly, the goal is to capture the essence of the problem. That this is a very challenging task is well illustrated by the long strand of literature on portfolio optimization, starting with the seminal work of Markowitz (1952) on the mean-variance model. This model produces the optimal portfolio of risky assets for different investors given estimates of expected returns and covariances. Although optimal given the assumed model, the resulting portfolios perform badly in practice. A large number of model extensions have therefore been proposed in literature with the aim of improving real-world performance. In DeMiguel et al. (2009), the mean-variance model and many of its extensions are evaluated in out-of-sample tests, and they show that none of the tested portfolio optimization models consistently outperform the naive equally weighted portfolio which is used as a benchmark. Further, they conclude that there are ‘many miles to go’ before portfolio optimization models produce improved decisions in practice. A major problem with the application of the mean-variance model is the mean (and to second order covariance) estimates. In addition, the assumptions of normally distributed returns, the single-period set-up, and the absence of transaction costs are all significant inconsistencies with reality, which decrease the quality of the optimal decision. Which of these properties are essential to model depends on the specific application. As an example, for a portfolio optimization problem with frequent rebalancing, the modeling of transaction costs and multiple periods is essential, while this is not so important with infrequent rebalancing.

A key lesson to learn from the literature on stochastic optimization in finance is that extreme care is needed to obtain good decisions, both in terms of the formulation of optimization problems and in terms of the input provided for the optimization models. The goal of obtaining good decisions is further complicated by the fact that optimization models most often are present also in generating the input, for example through statistical estimates or calibration of models to market data. The failure in capturing the essence of a problem can come from any of all these modeling choices, which together underlie the decisions suggested by the mathematical model. It is no doubt that stochastic optimization has the potential to make significant and important contributions to the understanding of financial markets and particularly for improved decision making, but significant challenges exist in achieving this.

1.2 Purpose and Delimitations

A consciousness of the challenges of applying optimization models to financial markets characterizes this thesis. The appended studies all aim to make contributions that support the process of improving decisions on financial markets, with particular attention paid to the potential importance of modeling details. The overall theme of this thesis is summarized by the following purpose.

The purpose is to develop and evaluate stochastic optimization models that support improved decisions under uncertainty on financial markets.

Decision making under uncertainty on financial markets covers a wide range of topics, which make delimitations necessary. In terms of stochastic optimization methods, this thesis focuses on stochastic programming and approximate dynamic programming. These modeling choices are motivated by the properties of the problems that we study, and particularly by the aim to capture many real-world complexities (see discussion in chapters 2 and 3). Further, we focus on decision problems covering financial risk management and portfolio optimization. The four studies appended in this thesis focus on, but are not limited to, these decision problems. Specifically, we study a risk management case for a global company; we investigate empirically a key modeling choice in stochastic programming; we suggest a method that can be used to efficiently model uncertainty in stochastic optimization; and we develop a model to solve a well-studied but, thus far, unsolved portfolio optimization problem. Although the thesis focuses on problems from financial markets, the contributions are not limited to these problems. In particular the methodological aspects covered here should also be relevant for decision problems in other fields. With the background and purpose presented, an overview on the structure of the thesis is now provided.

1.3 Outline

This is a compilation thesis composed of two main parts. The first part serves to give an overview of the thesis and the second part consists of the four appended papers. The main purpose of the first part is to show how the different studies are connected and related to the overall purpose of the thesis. Further, this part aims to motivate the different studies in relation to existing literature.

In chapter 2, an overview of commonly used methods for stochastic optimization is provided. The purpose of this chapter is partly to position the thesis in literature and partly to discuss the properties of the different stochastic optimization methods. Importantly, these properties underlie the methodological choices made in this thesis. Given this background, we are in chapter 3 ready to discuss the content, research objectives, results, contributions and further challenges of the four studies that constitute the foundation of the thesis. We end the first part in chapter 4 by returning to the overall purpose of the thesis and discuss to what extent this has been achieved. Further, we discuss the main contributions of the thesis and reflect on important methodological choices for obtaining the goal of improved (real-world) decisions. The second part of the thesis contains the four appended papers.

Methods for Stochastic Optimization

The purpose presented in the previous chapter emphasized that stochastic optimization models lie at the heart of this thesis. So far, however, little has been said about the methods used for stochastic optimization. Natural questions are: what is meant by a stochastic optimization model; how are these formulated; and which problem types are studied in the literature? These are the types of questions that this chapter aims to answer. As the area of stochastic optimization is both broad and deep, this presentation can not do it the justice of presenting the full richness of all its sub-fields. Instead, the aim of this chapter is to provide a brief overview which helps the reader gain intuition about stochastic optimization, and importantly, to position and motivate the studies making up this thesis. Motivated by the chosen modeling approaches in the appended studies, the presentation below will focus primarily on stochastic optimization models formulated in discrete time. For the interested reader, detailed treatments of different sub-fields of stochastic optimization (including continuous time modeling) can be found in for example Puterman (2005); Bertsekas (2017, 2012); Kirk (2004); Powell (2011); Bertsekas and Tsitsiklis (1996); Birge and Loveaux (2011); Ben-Tal et al. (2009).

To gain intuition about stochastic optimization, I believe it is helpful to start with a simple and illustrative example. Assume that you are given 100 SEK, and that you are offered to repeatedly bet on a fair coin (i.e., probability 50% for head/tail) to either double (*head*) or lose (*tail*) your entire capital. To incentivize betting, you are given 10 SEK every time you agree to gamble. Letting S_τ and S_τ^x denote the capital at time τ before and after the decision respectively, the problem can be represented by a decision tree as in Figure 1. While only two periods are shown, you may think about this problem as continuing over $T > 2$ (or even infinitely many) periods.

There is no single optimal strategy for this decision problem, since individuals

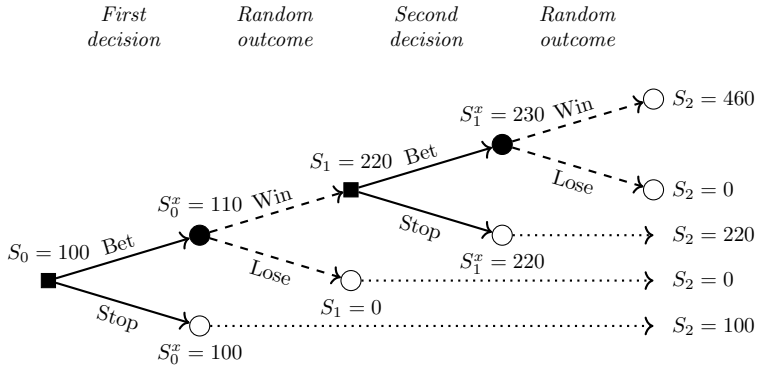


Figure 1: Decision tree example.

have different preferences for taking risks. Someone that strongly dislikes risks might refuse to gamble and take the 100 SEK, while a less risk-averse individual may decide to gamble up to a capital threshold of $K > 100$ SEK. Hence, to solve for the optimal decision(s) with a stochastic optimization model, we need to quantify an objective function capturing the preferences of the decision maker. The objective function could for example be to maximize the expected utility of the (random) capital at the horizon T , with utility described by a concave function capturing the degree of unwillingness to take on risks. An optimal strategy maximizes the objective function and returns a decision in each node of the tree that depends on the level of capital, S_τ which more generally describes the state of the system. Note that each decision is followed by a random outcome and that the state of the system evolves as¹

$$S_{\tau+1} = \begin{cases} 2(S_\tau + 10) & \text{if Bet and Win,} \\ 0 & \text{if Bet and Lose,} \\ S_\tau & \text{if Stop.} \end{cases}$$

While the example in Figure 1 is simplistic, it illustrates well the principal ingredients of stochastic optimization problems: an objective function (quantification of preferences); decisions (bet or stop); state of the system (the capital level); randomness (head or tail); and state dynamics (how capital evolves over time as a function of the previous capital level, decisions and the random outcomes). These five elements are proposed by Powell (2014, 2016, 2018) to represent the canonical form for sequential decision problems (see also e.g., Puterman, 2005). For the purpose of the presentation below, we introduce the following notation describing the five elements.

- State variables, S_τ , (and post-decision states S_τ^x);

¹Similarly, we can write down the dynamics of the post-decision state, S_τ^x , which is useful in some applications.

-
- Decisions x_τ , typically from a policy $X_\tau^\pi(S_\tau)$ that maps states into decisions, based on the allowed functions $\pi \in \Pi$;
 - Random (exogenous) information, $\xi_{\tau+1}$, that becomes known at $\tau + 1$;
 - State dynamics, $S_{\tau+1} = S^M(S_\tau, x_\tau, \xi_{\tau+1})$, describing how the system evolves;
 - Objective function, expressed in terms of a sequence of contributions (or costs) $C_\tau(S_\tau, x_\tau)$ implied by the dynamic system. Common objectives maximize (minimize) expected contributions (costs) or minimize a risk measure.

A broad class of stochastic optimization problems can be expressed in expectation form, and can be generalized by

$$\max_{\pi \in \Pi} \mathbb{E}^\pi \left[\sum_{\tau=0}^T C_\tau(S_\tau, X_\tau^\pi) | S_0 \right]. \quad (1)$$

Minimization of a risk measure, ρ , provides another significant avenue for stochastic optimization, and can be generalized by

$$\min_{\pi \in \Pi} \rho(C_0(S_0, X_0^\pi), \dots, C_T(S_T, X_T^\pi)). \quad (2)$$

While these two forms capture the majority of problem formulations in discrete time stochastic optimization, there are exceptions to the rule (robust optimization being one of them). Despite the seemingly generic formulations in (1) and (2), stochastic optimization problems come in a variety of forms. To see this, note that the five elements presented above, can each be of many different types, which combined produce a huge set of problem formulations. Key distinctions are based on the nature of the decision and state variables (discrete, continuous, low or high-dimensional).² Further, the state dynamics (linear, nonlinear, unknown), the type of randomness and the objective function underlie different sub-fields of stochastic optimization. Which road to follow is partly problem-driven and partly a modeling choice.

Below, we provide an overview of different approaches to study stochastic optimization problems based on a categorization into five sub-fields.³ These constitute partly distinct communities, with modeling styles and solution approaches originating from different classes of problems. While significant differences between these five groups exist, there are also important similarities and overlaps among them. One example is the approach of simplifying a complicated problem

²Note, for example, the limits of the decision-tree-approach presented in Figure 1 with respect to e.g., dimensionality.

³See e.g., Powell (2018) for a more comprehensive overview of the different sub-fields of stochastic optimization, including also e.g., reinforcement learning, optimal stopping, decision trees, and stochastic search

by breaking it into a set of easier-to-solve sub-problems in a recursive manner. This approach is utilized in many sub-fields of stochastic optimization, and has its origins in dynamic programming which can be considered fundamental to the study of sequential decision problems. Hence, it is natural to start an overview with this sub-field.

2.1 Dynamic Programming and Markov Decision Processes

While a dynamic program refers to a general sequential decision problem, the term dynamic programming is often used to describe a class of algorithms which are based on the principle of optimality. This principle is often associated with the work of Bellman (1957):

An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision. (Bellman, 1957, Ch. III.3)

For sequential decision problems that obey the principle of optimality, the search for an optimal policy function can be simplified into a set of optimization problems over decisions in a recursive manner. This is summarized by Bellman's optimality equations,

$$V_\tau(S_\tau) = \max_{x_\tau} (C_\tau(S_\tau, x_\tau) + E_\tau [V_{\tau+1}(S_{\tau+1})]), \quad (3)$$

where $V_\tau(S_\tau)$, referred to as the value function, represents the value of being in state S_τ given optimal decisions in subsequent periods. Underlying this representation is a Markovian structure⁴, which allows history to be represented by the state variable S_τ . A class of models for which dynamic programming is particularly well-suited falls under the category of Markov decision processes, which (typically) assume finite state and decision spaces. In this setting, the expectation of the next-period value function simplifies to a summation over all future states, $s' \in \mathcal{S}$, weighted by the so called transition probabilities, $p(s'|s, x_\tau)$, and actions a are chosen from a finite set $\mathcal{A}_s$. In this setting, Bellman's optimality equations can be expressed as (see e.g., Puterman, 2005)

$$V_\tau(s) = \max_{a \in \mathcal{A}_s} \left(C_\tau(s, a) + \sum_{s' \in \mathcal{S}} p(s'|s, a) V_{\tau+1}(s') \right). \quad (4)$$

The principle used in solving (4) is to compute $V_\tau(s)$ for all states $s \in \mathcal{S}$, based on stored values of $V_{\tau+1}(s')$ for all $s' \in \mathcal{S}$. This is similar in nature to a common approach used in solving decision trees, moving step-by-step from the leaves of

⁴Note that a Markovian structure can be constructed by extending the state vector.

the tree to the root node. In contrast, however, the value function recursion in (4) efficiently utilizes the concept of a state that determines decisions, which avoids (potentially) repeated calculations over identical information. While this extends the class of sequential decision problems that can be solved, computational limitations restrict the approach to enumerate states and decisions to small problems, i.e., problems with low dimension of the state and decision vector. Further, unless the transition probabilities are explicitly given (which is commonly not the case), the dimension of the random vector produces a third issue of dimensionality. Together, these are referred to as the curses of dimensionality, and are well known to restrict the class of problems that can be solved with dynamic programming. While these are serious limitations, it should be emphasized that the literature on dynamic programming (and Markov decision processes) is rich in both theoretical contributions underpinning stochastic optimization and examples of problems that do not suffer from the curses of dimensionality (see e.g., Puterman, 2005; Bertsekas, 2017).

The algorithmic strategy described by (4) is widely used to solve sequential decision problems in economics and finance. Closely related to this thesis is the classic dynamic portfolio choice problem of an investor with constant relative risk aversion (CRRA) preferences, which has been studied in a series of papers through the application of dynamic programming. The decision problem faced by the investor is to choose how much wealth to allocate between a set of risky assets and the risk-free asset, so as to maximize the expected utility of terminal wealth.⁵ The need for a dynamic formulation of the problem arises for example when transaction costs or predictability in returns are introduced. The case with predictable returns (and parameter uncertainty) is studied in Barberis (2000), and the combination of transaction costs and predictable returns is studied in Balduzzi and Lynch (1999), Lynch (2001), and Lynch and Tan (2010). In contrast to (4), both the state and the decision variables are continuous in these different instances of the portfolio choice problem. Further, randomness is driven by Gaussian noise, which does not obey a representation in terms of transition probabilities. The standard approach to making these problem instances fit into the form given in (4) is to construct discretized versions of the problems. In particular, the state (and decision) variables are discretized (e.g., on an equidistant grid) and randomness can be discretized for example by simulation (Barberis, 2000) or through a Gaussian quadrature approach (Balduzzi and Lynch, 1999; Lynch and Balduzzi, 2000; Lynch and Tan, 2010).⁶ These approximations lead to the idealized environment in (4), for which the problem can be solved by enumeration. While this approach can produce close-to-optimal solutions in these settings, it is limited by the curses of dimensionality. For this reason, the studies mentioned above are restricted to a single risky asset (and one predictive variable) or at most two risky assets (Lynch and Tan, 2010). From a financial perspective, we would also, of course, like to solve and under-

⁵The problem can be formulated to also capture the choice of intermediate consumption, see e.g., Lynch (2001).

⁶See e.g., Judd (1998) for an overview of different numerical integration methods.

stand optimal portfolio choice with more than two risky assets, and potentially in the presence of additional predictor variables. For this purpose, dynamic programming does not seem to offer a viable road.

We turn next to the related field of optimal control, where the focus is primarily on problems with continuous states and decision variables, often in a continuous time setting.

2.2 Optimal Control

As mentioned above, different sub-fields of stochastic optimization overlap to a large extent, and this is particularly true for optimal control, which is closely related to dynamic programming and Markov decision processes as well as for example reinforcement learning and approximate dynamic programming. The label that is used to describe a certain modeling approach is likely to depend on from which area one comes (computer science, engineering, operations research). Optimal control is tightly connected to the area of control theory and applications from engineering control problems such as robot navigation, auto-pilots, satellite navigation systems and control systems in space flights.

A broad class of optimal control problems are formulated in continuous time, using continuous controls, in a deterministic setting. The canonical stochastic optimal control problem is a continuous time problem with state dynamics governed by stochastic differential equations. Problems in this class are typically very hard to solve, and only a small number of stochastic optimal control problems obey a known analytical solution. Among the problems that have been solved, one of the best known is the linear quadratic regulator (LQR) problem. In a discrete time setting, this problem can be represented as in (1) with contribution functions, C_τ , that are quadratic in the states, S_τ , and controls (decisions), x_τ , and with state dynamics, S_τ^M that are linear in states and controls. The tractability of this problem has inspired researchers in finance to formulate dynamic portfolio choice problems that obey this special structure. Two recent examples of this are Gârleanu and Pedersen (2013) and Mei et al. (2016), who formulate dynamic mean-variance models with transaction costs in discrete time that correspond to the LQR problem.

There are also numerous studies in financial literature falling under the umbrella of continuous time stochastic optimal control. Closely related to this thesis is the problem of optimal portfolio choice in the standard setting with CRRA preferences, which has attracted considerable interest. In a series of papers on optimal portfolio (and consumption) choice with and without predictable returns, Merton (1969, 1971, 1973) derived the structural form of the optimal policy. These studies have later been extended in several dimensions. Of particular interest here is the introduction of transaction costs, which have been studied in for example Magill and Constantinides (1976); Davis and Norman (1990); Liu (2004); Muthuraman and Kumar (2006).

The studies mentioned above illustrate two different paths that both represent a successful application of optimal control to improve the understanding of financial markets. These are: (i) to formulate the problem in one of a small number of tractable forms (such as the LQR problem); or (ii) to study low-dimensional problems (typically 1-2 risky assets/state variables) with randomness and state dynamics in tractable forms. Both of these approaches have proven successful in understanding properties like hedging demand (Merton, 1971), the existence of a no-trade region (Magill and Constantinides, 1976; Davis and Norman, 1990), and the optimal policy given a specific class of investor preferences (Liu, 2004). However, it is also apparent that the standard tools of optimal control (in particular continuous time stochastic optimal control) are not well-suited to problems with high-dimensional state variables, non-standard randomness (non-Gaussian) and market frictions, particularly if these are combined in the same problem. An approach designed to handle (some of) the drawbacks of optimal control is presented next.

2.3 Approximate Dynamic Programming

Approximate dynamic programming (ADP) can be described as a class of methods that aim to handle problems where dynamic programming fails due to the curses of dimensionality. As such, ADP can be considered as a set of algorithmic strategies to solve large-scale dynamic programs. A commonly used trick to overcome the curses of dimensionality is to introduce value function approximations, $\bar{V}_\tau(S_\tau)$, based on estimates of the value, $v_\tau(S_\tau)$, in a subset of all states. These are related as (compare with (3))

$$v_\tau(S_\tau) = \max_{x_\tau} (C_\tau(S_\tau, x_\tau) + \mathbb{E}_\tau [\bar{V}_{\tau+1}(S_{\tau+1})]) . \quad (5)$$

Two principally different approaches for constructing the value function approximations from (5) are known as backward and forward approximate dynamic programming. The principle of the backward approach is to recursively construct value function approximations, $\bar{V}_\tau$, based on estimates of the value $v_\tau(S_\tau)$ in a set of sampled states. A popular approach for estimating $\bar{V}_\tau$ is through regression on a set of basis functions. The forward approximate dynamic programming approach starts from initial approximations of the value functions and updates these by stepping forward in time over sampled trajectories of random outcomes. Expressed differently, these algorithms learn the value of different states based on (5) by visiting random sequences of states over a large number of sample trajectories (see e.g., Powell, 2011, which discuss ADP based on this approach).

As an alternative to value function approximations, it is also common to construct ADP algorithms based on policy function approximations. A policy function approximation can for example be based on a polynomial function of the state variables. The use of neural networks as a way to approximate policy (or

value) functions has also attracted considerable interest, and links ADP to the broad area of artificial intelligence (see e.g., Bertsekas and Tsitsiklis, 1996).

The introduction of value and policy function approximations into the field of stochastic optimization opens up a potential to (approximately) solve problems that have previously been considered unsolvable (e.g., due to dimension). While several successful applications of ADP can be found in literature, the task of constructing a properly designed approximation is, however, generally very challenging. The key to making an ADP algorithm produce close-to-optimal solutions is to take advantage of problem structure, and this often requires detailed understanding of the problem under study (see e.g, Powell, 2011, for further discussion on these issues).

Several studies on ADP in financial literature have investigated dynamic portfolio choice in discrete time. An approach based on policy function approximations to study problems with predictable returns and parameter uncertainty is presented in Brandt et al. (2005). This approach is compared to value function approximations in van Binsbergen and Brandt (2007) and Garlappi and Skoulakis (2009). Approximate dynamic programming has also been applied to price American options (see e.g., Tsitsiklis and Van Roy, 2001; Longstaff and Schwartz, 2001). The approximate nature of ADP raises the issue of solution quality, and consequently, a need for methods to bound the distance to optimality. The key tools for this purpose are from duality theory, which has been extended into stochastic dynamic settings by for example (Rogers, 2002; Andersen and Broadie, 2004; Haugh et al., 2006; Haugh and Kogan, 2007; Brown et al., 2010; Brown and Smith, 2011).

We turn next to a field that does not (so much) rely on Bellman’s optimality equations, but instead utilizes the fact that stochastic optimization problems can be formulated as deterministic counterparts through a certain class of approximations.

2.4 Stochastic Programming

Different sub-fields of stochastic optimization can be distinguished partly by the approximations that are used to determine a solution (policy), and stochastic programming (SP) is characterized by an approximation strategy that discretizes the stochastic process governing the dynamics of the problem. The standard approach for this purpose is to construct a scenario-tree representation, and determine an approximate strategy based on this. Another distinguishing feature of SP is the tight link to mathematical programming, and in particular the use of convexity and duality theory from the rich field of deterministic optimization. A key reason for this intimate connection is that scenario-tree representations give rise to (large-scale) deterministic problems, referred to as deterministic equivalents. A large body of literature in SP describes algorithms

that utilize the structure of these problems, and these developments have made many large-scale problems resulting from SP tractable. Popular methods include Benders decomposition (see e.g., Van Slyke and Wets, 1969; Birge, 1985) and interior point methods (see e.g., Blomvall and Lindberg, 2002; Steinbach, 2003; Gondzio and Grothey, 2007).

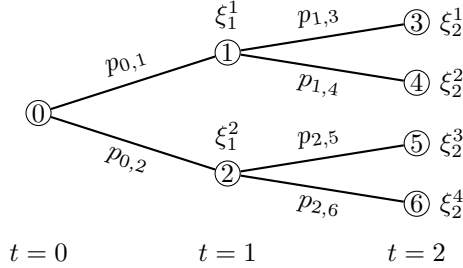


Figure 2: Illustration of scenario tree. Nodes represent information states and arcs realizations of randomness with associated conditional probabilities.

To illustrate the scenario-tree approach, Figure 2 presents an example of a scenario-tree. Given a random process ξ_τ with finite support, a scenario tree can be an exact representation, but most commonly it is an approximation based for example on Monte Carlo simulation. The principle behind constructing a scenario-tree through sampling is to generate a set of outcomes for the random variables in each non-leaf node (i.e., node 0,1 and 2 in Figure 2). Each random variable is governed by a conditional probability distribution, and an outcome from node i to j has probability $p_{i,j}$ to be realized. Further, every leaf-node (i.e., node 3-6) is associated with a scenario and the probability for each of these is given by the product of conditional probabilities along the respective scenario path. As an example, the scenario associated with node 3 has probability $p_{0,1} \cdot p_{1,3}$ to be realized.

Given that decisions are made at all three times in Figure 2, the model is typically referred to as a three-stage model with recourse, where the term recourse refers to a final (corrective) decision made after uncertainty is revealed (see e.g., King and Wallace, 2012). An important class of models in SP is two-stage models with recourse, which have been studied extensively in literature. While multi-stage models with (in theory) any number of stages can be constructed by the principle described above, the number of scenarios grows exponentially with the number of stages (and parameters), which in literature is referred to as the exponential growth of the scenario tree. Note that while dynamic programming is limited by the dimensions of the state, decision and randomness spaces, SP is limited mainly by the number of stages. Further, in contrast to the previously discussed sub-fields of stochastic optimization which produce an optimal policy or control, the (practically useful) output from an SP solution is the here-and-now decision⁷, i.e., the decision in node 0 in Figure 2. In practical

⁷While the optimal solution of an SP model can be viewed as a policy, note that this is

decision making, the idea underlying the SP approach is that a new model is constructed and solved every time a decision should be made.

A large class of SP models can be written in the form described by (1) (or (2)), with the expectation replaced by a sample counterpart. The idea of studying discretized versions of the ‘true’ models is known as the sample average approximation (SAA) method (see e.g., Kleywegt et al., 2001). For the purpose of solving these models with the tools of mathematical programming, a standard trick is to construct a deterministic formulation (deterministic equivalent) through what can be described as scenario-wise optimization. Using Figure 2 as an example, we can treat the scenario tree as four distinct scenarios with decision variables associated with, in total, 12 nodes. As this would imply perfect foresight of future information, so called non-anticipativity constraints are added to the model. These ensure that decisions corresponding to the same nodes in Figure 2 are identical (see e.g., Shapiro et al., 2014, ch. 3.1.5).

In summary, the scenario-tree approach of SP allows many otherwise intractable problems to be (approximately) solved. Particular advantages of SP as a modeling approach are its flexibility in handling general constraints and probability distributions, and the fact that it allows problems with state vectors of high dimensions to be (approximately) solved. Further, the reliance on simulation (SAA), makes SP amenable to statistical inference on for example the optimality gap of candidate solutions (see e.g., Shapiro et al., 2014). On the negative side is the exponential growth of the scenario-tree as the number of stages increase, and associated with this; potentially poor solution quality from scenario-trees of limited size.

Literature on SP closely related to this thesis are studies on scenario generation (Pflug, 2001; Dupačová et al., 2003; Høyland and Wallace, 2001; Høyland et al., 2003), hedging of currency risk (Volosov et al., 2005), and portfolio choice (ALM) problems (Carino et al., 1994; Consigli and Dempster, 1998; Kouwenberg, 2001; Fleten et al., 2002; Topaloglou et al., 2008; Ferstl and Weissensteiner, 2011). Other financial topics that have attracted interest include derivatives pricing (King, 2002; Pennanen and King, 2004; Topaloglou et al., 2008b), pricing problems in electricity markets (Pflug and Broussev, 2009; Kovacevic and Pflug, 2014), optimal choice of mortgage loans (Rasmussen and Clausen, 2007; Nielsen and Poulsen, 2004), and money management (Golub et al., 1995).

Next, we provide a brief overview of a field that is distinguished by the way it treats uncertainty.

restricted to the states captured by the nodes of the scenario-tree, which are generally very unlikely to be realized in practice.

2.5 Robust Optimization

While robust optimization is presented here as a sub-field of stochastic optimization, in literature it is often treated as a distinct field. The reason for this is that robust optimization takes a fundamentally different approach to uncertainty than is normally the case in stochastic optimization. Specifically, robust optimization represents uncertainty in terms of sets of possible outcomes without specifying associated probabilities. The rationale for this approach is to determine decisions that perform well under all (modeled) states of nature, making probabilistic specifications redundant. Classic applications for robust optimization are from engineering, where it is critical to maintain system performance. For the decision-maker, a robust decision means being immunized against uncertainty, which stands in contrast to a decision based on for example (1), as this is designed to perform well on average. The main paradigm in robust optimization is based on worst-case analysis, and the generic problem formulation can be written as

$$\max_{x \in X} \min_{W \in \mathcal{W}} f(x, W), \quad (6)$$

where $x \in \mathbb{R}^n$ denotes decisions, $X \subseteq \mathbb{R}^n$ the feasible domain, and $W \in \mathbb{R}^k$ random parameters taking values in the uncertainty set $\mathcal{W} \subseteq \mathbb{R}^k$. While not explicit in (6), the uncertainty can also impact the feasibility of the problem, i.e., $X(W)$.

A central question in the literature of robust optimization is tractability, i.e., (loosely speaking) which problem formulations are practically solvable. Generally, the introduction of uncertainty sets adds significant complexity in relation to a ‘standard’ deterministic problem. Tractability hinges (obviously) on the properties of the objective function, the feasible domain and the uncertainty set. For linear programs, a number of classes of uncertainty sets have been shown to produce tractable formulations, while nonlinear optimization have been less widely studied and understood (see e.g., Bertsimas et al., 2011; Gabrel et al., 2014). For reasons of tractability, the literature on robust optimization is largely focused on static (single-period) linear problems. Some examples of dynamic models can, however, be found in the literature, and are referred to as adjustable robust optimization models. Multiple stages add significant complexity to the problem, and only a few specialized formulations have been shown to be tractable (see e.g., Ben-Tal and Nemirovski, 2008).

A stream of literature related to this thesis studies portfolio choice based on robust optimization. As discussed in section 1.1, parameter uncertainty plays an important role for the poor practical performance of the mean-variance model. As a way to handle this, a number of papers have suggested robust versions of the mean-variance model, where the mean vector and the covariance matrix are assumed to lie in an uncertainty set (see e.g., Goldfarb and Iyengar, 2003; Garlappi et al., 2007). In a related study, Olivares-Nadal and DeMiguel (2018)

add transaction costs into the mean-variance model and formulates this as a robust optimization problem.

In summary, robust optimization provides a natural modeling choice when an inherently robust solution is sought (e.g., construction of a bridge) and it is also commonly applied when uncertainty in parameters are likely to have significant impact on the feasibility or sub-optimality (e.g., mean-variance) of a problem. However, the complexity of robust optimization limits the set of practically solvable models to a relatively small class of problems (mainly static-linear). Further, decisions in financial markets are generally about determining an optimal trade-off between risk and reward (expected return), and the worst-case nature of robust optimization is not, in its general form, designed for this. As a way to handle the issue of solutions being over-conservative, different ways to design the uncertainty set is studied in literature. However, more needs to be done to understand the economic rationales for robust solutions in many problems of financial economics.

Contribution of Thesis

In this chapter, we focus on the main contribution of the thesis; namely the four appended papers. Each paper aims to contribute to the overall purpose of this thesis by adding different pieces of important knowledge to support improved decisions on financial markets. A number of related topics are considered. Paper I studies the important question of how firms should hedge, and we find that properties inherent in this problem make it suitable to be studied with stochastic programming. Key issues for improving decisions of the model proposed in paper I, and stochastic programming models in general, are investigated in papers II and III. Specifically, paper II studies the important question of how more stages impact solution quality in stochastic programming and paper III develops an efficient scenario generation technique that allows scenario-trees of reduced size while maintaining the solution quality. Despite access to an efficient scenario generation technique, the stochastic programming approach is not well suited for problems with a large number of decision stages, due to the exponential growth of the scenario tree. Motivated by this methodological limitation, paper IV investigates how approximate dynamic programming can be used to study problems with a large number of decision stages (and many state variables). Specifically, we study how approximate dynamic programming can be used to solve a classic portfolio choice problem, which despite decades of interest from researchers in financial literature has remained unsolved given more than a few risky assets.

3.1 Paper I

The decision problem studied in paper I has its origin in a real case faced by a collaboration company, but is, in its generic form, of high relevance for most global firms; namely the hedging of currency and interest rate risk resulting from a set of (potentially uncertain) cash flows. The decision maker, who we call the risk manager, faces the problem of deciding first of all whether to hedge,

and if so, how to set up a good hedge. In some cases operational hedging¹ may provide a good alternative, but in most cases, financial hedging is needed. In the latter case, the decision problem is basically which type and volume of financial instruments to enter into. While it may seem to be a problem of limited complexity, we must take into account that the menu of relevant financial instruments to choose from is large and that the set of cash flows generally span over multiple currencies and many points in time. The interaction of currency, interest rate and cash flow risks creates a problem that, with the exception of cases including only a few cash flows, is too complex to be overseen and well understood without mathematical support. Further, it is far from obvious what defines a good hedge in this setting, that is whether we should aim to minimize for example a measure of variance or a risk measure like expected shortfall. Associated with this is the cost of hedging, or more generally the trade-off between risk and costs, which is at the heart of financial decisions. To contrast the computational approach, two common alternatives in the industry are: (i) to leave the exposure unhedged; and (ii) to use a simple heuristic like cash flow matching. In the worst case, the choice to remain unhedged may jeopardize the existence of the whole company, while cash flow matching—if even possible—is likely to be very expensive, and so seriously impact the profitability of the firm.

The problem of corporate hedging has attracted considerable interest in financial literature, dating back (at least) to the seminal work of Modigliani and Miller (1958). Their work basically answers the question of why firms should hedge, and this answer is summarized by the existence of financial frictions. In relation to the well studied ‘why’ question, much less is known about how firms should hedge. While some results are available for simplified settings, such as a single cash flow (see e.g. Ederington, 1979; Kerkvliet and Moffett, 1991), the complex problem faced by the risk manager is far from being well understood. The real-world problem contains a number of fundamental characteristics not captured in existing models, such as non-normally distributed foreign exchange rates, time-varying term premia in interest rate markets², transaction costs, trading at market prices, and not least, access to the rich asset universe available in financial markets. To support improved decisions, it is likely that it is important to model (at least some of) these characteristics. In other words, we expect some of these to be critical for capturing the essence of the real problem. To enable the inclusion of these types of complexities, we work in a stochastic programming framework which in contrast to for example dynamic programming and optimal control allows the combination of ‘non-standard’ distributions, transaction costs, trading at market prices and a large state space. As highlighted in section 2.4, the flexibility of the stochastic programming approach comes, however, at the cost of a limitation in the number of stages, and this confines the possibility to fully capture the dynamic nature of the problem. In the paper, we choose to

¹ An example of operational hedging is to locate production in the market (currency) where cash flows are generated.

² Term premia refer to a compensation for risk in instruments (e.g., bonds) with longer time to maturity. As a consequence, hedging instruments have different expected returns.

formulate the optimization problem as a two-stage model, although the problem is dynamic in nature.

Motivated by the identified gap in literature and the properties of the decision problem, the overall research objective of this paper was to develop a stochastic programming framework that can be used to support real-world corporate hedging. Using this framework, a number of interesting questions could be investigated. First, we aimed to study the implications of different objectives on the structure of the optimal hedge portfolio and the expected cost of the hedge. Further, we aimed to investigate the importance of modeling a rich universe of hedging instruments, with respect to being able to reduce risk and to set up cost-efficient hedges in particular.

In the paper, we study a problem instance with annual uncertain cash flows up to ten years in four markets (currencies), and the optimization problem is that of selecting a hedging portfolio among (up to) 66 annually spaced currency forward contracts and interest rate swaps. The results of this paper show important differences in the properties of the hedge when variance instead of expected shortfall is minimized. Specifically, variance minimization implies trading in many more assets and to higher expected costs. Further, the results highlight that a rich-enough asset universe is important not only for the decision maker's potential to control risk, but also for the expected cost of the hedge. These findings are tightly linked to the modeling of term premia (and transaction costs), as this allows us to capture the distinct risk-return profile of different hedging instruments. The insights into the impact of the objective function and the importance of the asset universe and term premia are key contributions of this paper. It is worth noting that these findings are made possible by the development of the stochastic programming framework for corporate hedging. This framework, along with all its components, is also an important contribution of this paper.

As the overall purpose of this thesis is focused on improving decisions, it is essential to reflect on potential areas for model improvement. A key lesson to learn from paper I is the importance of term premia for the properties of the model-implied hedge. For the purpose of modeling the well-documented term premia on interest rate markets, we worked with the affine term structure model of Duffee (2002). A serious drawback of this model is that the parameter estimation produces a non-convex optimization problem with many local optima. A critical issue is that we experienced different local optimal with significantly different economic implications. In the paper, we propose a way to handle this, but it does not completely resolve the issue of non-convexity. In the search for alternative models, it is important to note that the development of arbitrage-free term structure models with time-varying term premia is a challenging topic and that few models are available in the literature. To the best of our knowledge, no better models serving the needs of the proposed framework are currently available in the literature, but still, we stress that this is an important ingredient for obtaining improved decisions.

Another possibility for improved decisions is to extend the model into a multi-stage setting. Since the nature of the hedging problem is dynamic, in theory more stages are preferable. In practice, however, the price of adding stages to the model is a coarser discretization of randomness, and the optimal trade-off is far from obvious. This issue is carefully investigated in paper II, which aims to add insights into the value and cost of more stages in stochastic programming. Related to this question is the topic of scenario generation, which is the theme of paper III. Specifically, paper III suggests an efficient method for scenario generation, thereby allowing coarser discretizations with maintained solution quality, which paves the way for models with more stages.

3.2 Paper II

This paper studies an important modeling choice in stochastic programming, namely how many decision stages to model given a problem of inherently multi-stage nature. The key reason for the importance of this modeling choice is a computational limitation in terms of scenario tree size, which implies that more stages are associated with a coarser representation of uncertainty. Specifically, the exponential growth of the scenario tree in the number of stages quickly confines the possibility to accurately represent uncertainty when more stages are added to the model.³

While several studies in literature compare the performance of a multi-stage instance relative a static benchmark strategy (see e.g., Fleten et al., 2002; Kouwenberg, 2001; Topaloglou et al., 2008), to the best of our knowledge no study investigates the choice of the number of decision stages. This gap in the literature motivates the overall research objective of this paper, which is to advance the understanding on the impact of adding stages in stochastic programming models for problems of multi-stage nature.

To investigate the impact of more stages in stochastic programming models, we evaluate the solution quality resulting from models with different number of stages. We build the evaluation on three key ingredients. First, we can obtain pessimistic bounds by simulating stochastic programming strategies with here-and-now decisions implemented in sequence⁴. This provides an unbiased estimator of how the associated strategies perform on the original problem. Second, we can obtain optimistic bounds from the optimal objective function values of sample average approximations, which combined with the pessimistic bounds provide important information on solution quality. Finally, we can estimate optimality gaps implied by stochastic programming strategies for problem

³Note that for e.g., dynamic programming and optimal control, the decision structure is (generally) not an issue.

⁴This simulation approach has been suggested in e.g., Fleten et al. (2002) and Kouwenberg (2001).

instances that can be solved ‘exactly’ with other stochastic optimization approaches.

For the purpose of the numerical investigation, we study a classic portfolio choice problem in financial literature which allows the value of multi-stage modeling to come from different sources. Specifically, we study problem instances with proportional transaction costs and/or predictability in returns. These are practically and empirically well justified aspects of the portfolio choice problem faced by investors. Importantly, the different problem instances capture varying channels through which more stages create a (potential) value for the investor, which enrich the analysis on the impact of adding stages in stochastic programming.

As emphasized by Kaut et al. (2007), the solution quality implied by different scenario trees is problem dependent, and hence no single scenario generation method will produce the best solution for all problems. For the same reason, we can not find a generic answer to the question of the optimal number of decision stages. This does (obviously) depend on a number of factors such as the problem instance, the scenario generation method, the allowed problem size and the branching structure. Hence, the key insights to take from this study are not the specific numerical results but instead the generic properties of adding stages.

A number of fundamental properties associated with adding more stages in stochastic programming are highlighted through the numerical experiments. First, while more stages (naturally) can improve solution quality, the impact of a coarser representation of uncertainty can also be substantial. As a consequence, more stages can, but does not necessarily, produce improved solution quality. Second, the gap between pessimistic and optimistic bounds increases with the number of stages. This property makes stochastic programming models with many stages particularly vulnerable to the risk of overestimating performance, unless out-of-sample evaluation is also performed. Finally, the numerical experiments highlight the importance of understanding what drives the value of more decision stages, and particularly if this can be captured in models with only a few stages.

On a higher level, the results suggest to not blindly let the problem determine the number of modeled decision stages, but instead to let performance evaluation guide this choice. Further, the results are reminders of the approximate nature of stochastic programming, which is (primarily) used to support decisions for problems that lack ‘exact’ solution. The goal for the modeler is to design this approximation such that the impact on solution quality is minimized, and the results of this paper accentuate the importance of carefully choosing the number of decision stages while utilizing properties of the studied problem.

It is clear from this study that it can be challenging to realize the benefits of multi-stage models in practice. With the aim of developing models that support improved decisions, it is natural to investigate the key element for successful extension into multi-stage models, namely an efficient scenario generation method.

This is the topic of paper III, which more specifically suggests a scenario generation method based on importance sampling.

3.3 Paper III

This paper studies scenario generation for the purpose of sample average approximations in stochastic optimization, with particular focus on multi-stage stochastic programming. Specifically, we propose a scenario generation technique based on Monte Carlo simulation with importance sampling. The principal tool of importance sampling is a change of probability measure, and the challenge is to design this to produce a low sample variance. A central result that can guide the change of probability measure is the existence of a minimum-variance distribution which under certain conditions produces a sample estimator with zero variance. While this result relies on the true mean being known, it illustrates the potential embedded in importance sampling—if an approximation to the zero-variance distribution can be constructed, a substantial reduction in sample variance can be expected. Despite the high potential of importance sampling and the large interest of scenario generation in stochastic programming literature, only a few studies can be found on this topic, with Parpas et al. (2015) being a recent example.

The methodological focus on importance sampling in this study is a consequence of an issue investigated during the work with paper II, namely the problem of leveraged solutions from sample average approximations. Specifically, for the studied portfolio choice problem, sample average approximations tend to produce solutions that imply short-selling, which given the continuous (e.g., log-normal) distribution are infeasible.⁵ The reason underlying this is the lack of ‘extreme’ scenarios, and the idea was to use importance sampling to enable us to sample more heavily from these regions. While we did not completely resolve this issue (see discussion in the paper), the implementations showed us the potential of importance sampling for sample mean estimation. Specifically, given an initial simple and ‘rough’ method for constructing the importance sampling distribution, we documented sample mean estimators with lower variance than those produced by standard variance reduction methods. These findings, combined with the identified gap in literature, motivated a deeper investigation into the potential of importance sampling for stochastic optimization, and in particular, for stochastic programming. Taken together, these factors underlie the overall research objective of the paper, namely to investigate how importance sampling can be designed to produce high solution quality with few scenarios in stochastic optimization, and in multi-stage stochastic programming in particular. The problem that we focus on for this purpose is the classic portfolio

⁵The reason for the infeasibility is that short-selling implies negative wealth with positive probability.

choice problem with constant relative risk aversion preferences, proportional transaction costs and log-normal returns.

The main result of this paper is a method for constructing an analytical approximation of the zero-variance distribution derived for the studied portfolio choice problem. We show that this method produces significant improvements in sampling efficiency compared to standard methods, and, importantly, higher solution quality in stochastic optimization. The main contributions of this paper are the suggested approach to construct an importance sampling distribution and its application for the generation of scenario trees in multi-stage stochastic programming. Further, this paper makes a contribution to the stochastic optimization literature through a comprehensive evaluation of a number of commonly-used scenario generation (variance reduction) techniques.

Frankly, the documented performance of the proposed importance sampling method was surprising to us. That as few as ten scenarios based on importance sampling produce the same solution quality as thousands or even millions of scenarios from standard methods suggests a large potential for improving solutions from stochastic optimization models. While the proposed method may be applied in many areas of stochastic optimization, stochastic programming is of particular interest due to its reliance on sample average approximations. The results in this paper motivate further research into the application of importance sampling for stochastic optimization in general and stochastic programming in particular. To extend the method to a larger set of problems and study how it can be generalized to handle other distributions is of particular interest.

Improved sampling efficiency is particularly important when adding more stages to stochastic programming models. However, the scenario tree approach still limits the number of stages. A key reason for this in financial applications is arbitrage-free modeling, which requires at least as many outcomes per period as number of assets.⁶ Hence, for problems with many assets and a large number of time periods, stochastic programming can basically not be used due to the exponential growth of the scenario tree. In the fourth paper, we study a problem of this type, and thus we had to consider alternative methods.

3.4 Paper IV

In paper IV, we study a classic problem in financial literature, namely dynamic portfolio choice in the presence of transaction costs. The standard setting for this problem is an investor with constant relative risk aversion preferences, who wants to maximize the expected utility of terminal wealth. Both continuous and discrete time versions of this problem have attracted considerable interest in literature, and here we focus on the discrete time case. The key challenge in

⁶Note that constraints, such as restrictions on borrowing and short-selling, may hide these problems and that the solutions in such cases are likely to be irrelevant.

solving this problem is, given many risky assets, to handle the large state space resulting from the presence of transaction costs. Specifically, with transaction costs the value function depends on the investor's asset positions, and hence the state space grows with the number of risky assets.

This problem has attracted considerable interest in literature; in continuous time under the label of stochastic optimal control, and in discrete time based on the principles of dynamic programming. Both streams of literature have successfully solved the problem given one or at most a few risky assets. This restriction to a low number of assets is not theoretically or practically motivated, but is a critical limitation of these models. That the literature is striving for methods to solve problems with many risky assets is well summarized by Muthuraman and Kumar (2006) in their reflections on future research:

Arguably the most exciting direction would be using approximations with proper justifications and building a tool that can compute optimal policies for, say, a 10-stock problem. This would bring the literature in the area of portfolio optimization with transaction costs to a point where it can provide fund managers a tool that can be used for calibration and scientific decision making.
(Muthuraman and Kumar, 2006, p 334)

More than a decade later, to the best of our knowledge little progress had been made on this problem. As concluded above, the commonly used approaches of dynamic programming and optimal control are badly suited for problems with many risky assets, and stochastic programming is limited by the number of decision stages. Hence, the key to making progress seems to be to adopt other optimization methods. A fundamental property of any candidate method is to allow many assets, i.e., to overcome the curse of dimensionality. As suggested by Muthuraman and Kumar (2006), this inevitably requires the use of approximations which should be properly justified. Among the tools available for stochastic optimization, the approach of approximate dynamic programming (ADP) appeared to be best suited for the purpose of solving this problem.

The properties of the portfolio choice problem and, importantly, the gap in the literature, motivate the overall research objective of this paper, which was to study how ADP can be used to solve the classic dynamic portfolio choice problem with transaction costs to high accuracy, given many risky assets and a large number of time periods. The approximate nature of ADP inevitably raises the question of solution quality, i.e., how well the original problem is solved. Since the true optimum is unknown, we can not determine the optimality gap. However, we can obtain bounds on the distance to optimality through the use of duality theory.

In the numerical experiments, we study problem instances with ten risky assets and up to 52 periods. Motivated by the properties of the problem, we build the ADP model around a quadratic approximation of the value function in the

no-trade region, and we estimate these in a recursive manner following the principles of Bellman (1957). To evaluate the performance of the proposed model, we use a number of benchmark heuristics from literature, and we estimate optimistic bounds using the dual approach in Brown and Smith (2011). The results show that the ADP model consistently outperforms the benchmark heuristics, and we document duality gaps, expressed as annualized certainty equivalent returns, in the range 0.003-0.06%. The proposed ADP model that is shown to produce close-to-optimal strategies for the classic portfolio choice problem is the main contribution of this paper. Part of our findings is that the value function approximations of the ADP model can be used to obtain improved optimistic bounds when used with the dual method from Brown and Smith (2011).

We end this discussion with a reflection on potential areas of improvement and promising directions to extend the model. First, to maintain transparency for the readers of the paper, we have chosen to work with a well-known scenario generation technique for the purpose of sample mean approximations, namely Monte Carlo simulation with Latin hypercube sampling. To improve sampling efficiency, the importance sampling method proposed in paper III is a promising candidate. Further, little effort has been devoted to the question of how the sampling of states (portfolios) is made efficient, and so it is likely that this too can be improved.

There are several interesting directions along which this paper can be extended. A question of great interest in financial literature is how predictability in returns impact optimal portfolio choice. From a modeling perspective, predictability in returns implies one (or several) additional state variable(s), and in literature a common approach to model this is through a latent state variable model, based on for example the dividend-price ratio. To extend the model in this direction, a careful examination of how this state variable impacts the value function is needed.

This paper is focused on presenting the suggested ADP model and, importantly, to obtain bounds on the distance to optimality. With assurance of close-to-optimal strategies, a number of interesting financial questions for the case with many risky assets, such as properties of the no-trade region, and of optimal trading behavior, can be further analyzed.

Summary of Thesis

In this closing chapter to the first part, we return to the purpose of the thesis and discuss to what extent it has been achieved. Basically, we discuss to what extent this thesis supports improved decisions under uncertainty in financial markets. We end with a discussion about the process of improving decisions for financial markets through the use of stochastic optimization.

4.1 Conclusions

Let us start by reiterating the purpose of the thesis:

The purpose is to develop and evaluate stochastic optimization models that support improved decisions under uncertainty on financial markets.

Aligned with the purpose, the four studies that comprise this thesis all investigate decision making under uncertainty by developing and/or evaluating stochastic optimization models, with an emphasis on financial applications. Three of the four papers focus on the development of stochastic optimization models. Specifically, paper I develops a stochastic programming framework for corporate hedging, paper III develops a scenario generation method based on importance sampling, and paper IV develops an ADP model designed for the classic portfolio choice problem with CRRA preferences and transaction costs. To support improved decisions, evaluation constitutes a central ingredient in all four studies. An important distinction should, however, be made regarding the nature of the different evaluations. Specifically, in paper III and IV, we study theoretical problems with known (assumed) distributions which enable evaluation through simulation, and, importantly, statistically significant improvements to be documented. Similarly, the evaluation of stochastic programming models with different number of stages in paper II is based on an assumed known distribution, which allow simulation-based comparisons. In contrast to these three

studies, the corporate hedging problem in paper I originates from a practical case with the ‘true’ distribution being unknown. As a consequence, the evaluation is based on out-of-sample tests on real data. While these tests produce insights into properties of the hedging portfolio and suggest the model to be robust in practice, the amount of data is insufficient to allow well-supported conclusions of decision quality.

The documented improvements for theoretical models are important steps in the process of improving decisions in practice, not least for allowing further issues to be better investigated. Taking the classic portfolio choice problem which we solve to high accuracy in paper IV as an example, the step into practice creates the need for parameter estimation (mean and covariance), and raises questions about the relevance of assuming for example log-normal returns, CRRA preferences, and proportional transaction costs¹. With the documented improvements for theoretical models, these types of real-world issues can be further investigated in the process of building scientifically based decision support. In conclusion, this thesis contributes with steps in the process of improving decisions on financial markets through the use of stochastic optimization models. The major contributions underlying the steps of improved decisions are:

- A stochastic programming framework for corporate hedging, which through proper modeling of many real-world complexities revealed new insights into the importance of term premia, the impact of the objective function, and the value of a rich-enough asset universe;
- A comprehensive evaluation on the impact of more stages in stochastic programming models, which gives insights into the challenges of realizing improved decisions by adding stages to stochastic programming models;
- A scenario generation method based on importance sampling that is shown to produce significantly improved sampling efficiency for the portfolio choice problem with CRRA preferences and log-normal returns;
- An ADP model that gives close-to-optimal solutions to the classic, and thus-far unsolved, portfolio choice problem with CRRA preferences and proportional transaction costs, given many risky assets and a large number of time periods.

Further, the thesis highlights the advantages and disadvantages of different approaches used to study stochastic optimization problems, and related to this—how the choice of approach impacts the ability to solve different problems. In particular, this thesis illustrates how problems, that due to several real-world complexities are badly suited for e.g., dynamic programming and optimal control, can be better studied with stochastic programming and approximate dynamic programming.

¹Proportional transaction costs does not capture potential market impact from trading, which is an issue particularly for illiquid assets, or if the trading volume is large.

4.2 Improving (real-world) Decisions

In this final section, we reflect on important methodological choices in stochastic optimization, and in particular their relation to the goal of improving (real-world) decisions. As a base for this discussion, we use the simple conceptual model in Figure 1, which illustrates the process from a real problem to a solution, through a (stochastic optimization) model of reality.

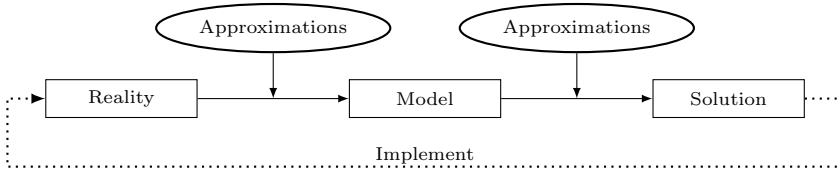


Figure 1: An illustration of the process from problem to solution.

We start with a decision problem of reality and the intended destination is a solution that will either be directly implemented or used as guidance. The overall aim is to provide solutions that support improved decisions, and a fundamental challenge in this process is the design of a model that, despite the impossibility of exactly representing reality, captures the essence of the problem. Hence, we have to live with the fact that the model from which we determine a solution is only an approximation of the problem that we would like to solve. In other words, approximations are inevitable in the step from reality to a model in Figure 1.

In contrast, approximations can be avoided in the step from model to solution. This hinges, however, on the model obeying a structure that admits an ‘exact’ solution. Traditionally, these are the types of model formulations that have attracted most interest in financial literature. Specifically, these models allow either closed-form or ‘exact’ numerical solutions, most commonly through the application of dynamic programming or optimal control. An obvious benefit of these models is that the properties of the solution can be traced back to the assumptions of the model, and in the ideally-sought case of closed-form solutions, valuable insights into structural properties of the solution may be obtained. However, the price to pay is (generally) serious limitations in the model formulations allowed, and thus also the ability to capture the essence of the real problem. Critically, the insights from an overly simplified model may even be misleading.

A central message of this thesis is that significant and important flexibility to reduce the impact of approximations in the step from reality to a model can be obtained by allowing approximate methods to solve the model. A well-known but often overseen point is that both steps of approximations jointly determine solution quality. Hence, the use of approximate methods to solve the model is simply justified by the aim being to reduce the effect of the total approximation.

This motivates the focus in this thesis on stochastic programming and approximate dynamic programming, which through different approximation strategies provide us the important flexibility necessary for studying additional aspects of reality.

It is worth emphasizing that the modeling of new aspects is a fundamental component in the process of improving decisions through stochastic optimization. Another fundamental component is a systematic evaluation of candidate models. The use of approximate methods to solve the model introduces an additional step of evaluation; namely how well the model is solved. The solution quality in this step can generally be well estimated, for example through pessimistic and optimistic bounds (as in papers II, III and IV). Given control of the solution quality in this step, the consequences of the model approximations can be properly studied through evaluation of decision quality in reality. The iterative procedure of modeling new aspects, solving these models, and evaluating the resulting decision quality describes the process of improving decisions through stochastic optimization.

Bibliography

- Andersen, L. and Broadie, M. (2004), ‘Primal-dual simulation algorithm for pricing multidimensional american options’, *Management Science* **50**(9), 1222–1234.
- Balduzzi, P. and Lynch, A. W. (1999), ‘Transaction costs and predictability: Some utility cost calculations’, *Journal of Financial Economics* **52**(1), 47–78.
- Barberis, N. (2000), ‘Investing for the long run when returns are predictable’, *The Journal of Finance* **55**(1), 225–264.
- Bellman, R. E. (1957), *Dynamic programming*, Princeton University Press, Princeton, New Jersey.
- Ben-Tal, A., El Ghaoui, L. and Nemirovski, A. (2009), *Robust optimization*, Princeton University Press.
- Ben-Tal, A. and Nemirovski, A. (2008), ‘Selected topics in robust convex optimization’, *Mathematical Programming* **112**(1), 125–158.
- Bertsekas, D. P. (2012), *Dynamic programming and optimal control: Approximate dynamic programming*, Vol. II, 4 edn, Athena Scientific.
- Bertsekas, D. P. (2017), *Dynamic programming and optimal control*, Vol. I, 4 edn, Athena Scientific.
- Bertsekas, D. P. and Tsitsiklis, J. N. (1996), *Neuro-Dynamic Programming*, 1st edn, Athena Scientific.
- Bertsimas, D., Brown, D. B. and Caramanis, C. (2011), ‘Theory and applications of robust optimization’, *SIAM review* **53**(3), 464–501.
- Birge, J. R. (1985), ‘Decomposition and partitioning methods for multistage stochastic linear programs’, *Operations Research* **33**(5), 989–1007.
- Birge, J. R. and Loveaux, F. (2011), *Introduction to Stochastic Programming*, 2 edn, Springer.

- Blomvall, J. and Lindberg, P. O. (2002), ‘A riccati-based primal interior point solver for multistage stochastic programming’, *European Journal of Operational Research* **143**(2), 452–461.
- Brandt, M. W., Goyal, A., Santa-Clara, P. and Stroud, J. R. (2005), ‘A simulation approach to dynamic portfolio choice with an application to learning about return predictability’, *The Review of Financial Studies* **18**(3), 831–873.
- Brown, D. B. and Smith, J. E. (2011), ‘Dynamic portfolio optimization with transaction costs: Heuristics and dual bounds’, *Management Science* **57**(10), 1752–1770.
- Brown, D. B., Smith, J. E. and Sun, P. (2010), ‘Information relaxations and duality in stochastic dynamic programs’, *Operations Research* **58**(4, part 1), 785–801.
- Carino, D. R., Kent, T., Myers, D. H., Stacy, C., Sylvanus, M., Turner, A. L., Watanabe, K. and Ziemba, W. T. (1994), ‘The russell-yasuda kasai model: An asset/liability model for a japanese insurance company using multistage stochastic programming’, *Interfaces* **24**(1), 29–49.
- Consigli, G. and Dempster, M. A. H. (1998), ‘Dynamic stochastic programming for asset-liability management’, *Annals of Operations Research* **81**(0), 131–162.
- Davis, M. H. and Norman, A. R. (1990), ‘Portfolio selection with transaction costs’, *Mathematics of Operations Research* **15**(4), 676–713.
- DeMiguel, V., Garlappi, L. and Uppal, R. (2009), ‘Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy?’, *Review of Financial Studies* **22**(5), 1915–1953.
- Duffee, G. R. (2002), ‘Term premia and interest rate forecasts in affine models’, *Journal of Finance* **57**, 405–443.
- Dupačová, J., Gröwe-Kuska, N. and Römisch, W. (2003), ‘Scenario reduction in stochastic programming: an approach using probability metrics’, *Mathematical programming* **95**(3), 493–511.
- Ederington, L. H. (1979), ‘The hedging performance of the new futures markets’, *Journal of Finance* **34**, 157–170.
- Ferstl, R. and Weissensteiner, A. (2011), ‘Asset-liability management under time-varying investment opportunities’, *Journal of Banking & Finance* **35**(1), 182–192.
- Fleten, S.-E., Høyland, K. and Wallace, S. W. (2002), ‘The performance of stochastic dynamic and fixed mix portfolio models’, *European Journal of Operational Research* **140**(1), 37–49.

- Gabrel, V., Murat, C. and Thiele, A. (2014), ‘Recent advances in robust optimization: An overview’, *European Journal of Operational Research* **235**(3), 471–483.
- Garlappi, L. and Skoulakis, G. (2009), ‘Numerical solutions to dynamic portfolio problems: The case for value function iteration using taylor approximation’, *Computational Economics* **33**(2), 193–207.
- Garlappi, L., Uppal, R. and Wang, T. (2007), ‘Portfolio selection with parameter and model uncertainty: A multi-prior approach’, *The Review of Financial Studies* **20**(1), 41–81.
- Gârleanu, N. and Pedersen, L. H. (2013), ‘Dynamic trading with predictable returns and transaction costs’, *The Journal of Finance* **68**(6), 2309–2340.
- Goldfarb, D. and Iyengar, G. (2003), ‘Robust portfolio selection problems’, *Mathematics of operations research* **28**(1), 1–38.
- Golub, B., Holmer, M., McKendall, R., Pohlman, L. and Zenios, S. A. (1995), ‘A stochastic programming model for money management’, *European Journal of Operational Research* **85**(2), 282–296.
- Gondzio, J. and Grothey, A. (2007), ‘Parallel interior-point solver for structured quadratic programs: Application to financial planning problems’, *Annals of Operations Research* **152**(1), 319–339.
- Haugh, M. B. and Kogan, L. (2007), *Duality Theory and Approximate Dynamic Programming for Pricing American Options and Portfolio Optimization*, Elsevier, chapter 22, pp. 925–948.
- Haugh, M. B., Kogan, L. and Wang, J. (2006), ‘Evaluating portfolio policies: A duality approach’, *Operations Research* **54**(3), 405–418.
- Høyland, K., Kaut, M. and Wallace, S. W. (2003), ‘A heuristic for moment-matching scenario generation’, *Computational Optimization and Applications* **24**(2), 169–185.
- Høyland, K. and Wallace, S. W. (2001), ‘Generating scenario trees for multistage decision problems’, *Management Science* **47**(2), 295–307.
- Judd, K. L. (1998), *Numerical methods in economics*, MIT Press, Cambridge, Massachusetts London, England.
- Kaut, M. and Wallace, S. W. (2007), ‘Evaluation of scenario-generation methods for stochastic programming’, *Pacific Journal of Optimization* **3**(2), 257–271.
- Kerkvliet, J. and Moffett, M. H. (1991), ‘The hedging of an uncertain future foreign currency cash flow’, *Journal of Financial and Quantitative Analysis* **26**, 565–578.

- King, A. J. (2002), ‘Duality and martingales: a stochastic programming perspective on contingent claims’, *Mathematical Programming* **91**(3), 543–562.
- King, A. J. and Wallace, S. W. (2012), *Modeling with stochastic programming*, Springer Science & Business Media.
- Kirk, D. E. (2004), *Optimal control theory: An introduction*, Dover.
- Kleywegt, A. J., Shapiro, A. and Homem-de Mello, T. (2001), ‘The sample average approximation method for stochastic discrete optimization’, *SIAM Journal on Optimization* **12**(2), 479–502.
- Kouwenberg, R. (2001), ‘Scenario generation and stochastic programming models for asset liability management’, *European Journal of Operational Research* **134**(2), 279–292.
- Kovacevic, R. M. and Pflug, G. C. (2014), ‘Electricity swing option pricing by stochastic bilevel optimization: a survey and new approaches’, *European Journal of Operational Research* **237**(2), 389–403.
- Liu, H. (2004), ‘Optimal consumption and investment with transaction costs and multiple risky assets’, *The Journal of Finance* **59**(1), 289–338.
- Longstaff, F. A. and Schwartz, E. S. (2001), ‘Valuing american options by simulation: a simple least-squares approach’, *The Review of Financial Studies* **14**(1), 113–147.
- Lynch, A. W. (2001), ‘Portfolio choice and equity characteristics: Characterizing the hedging demands induced by return predictability’, *Journal of Financial Economics* **62**(1), 67–130.
- Lynch, A. W. and Balduzzi, P. (2000), ‘Predictability and transaction costs: The impact on rebalancing rules and behavior’, *The Journal of Finance* **55**(5), 2285–2309.
- Lynch, A. W. and Tan, S. (2010), ‘Multiple risky assets, transaction costs, and return predictability: Allocation rules and implications for us investors’, *Journal of Financial and Quantitative Analysis* **45**(4), 1015–1053.
- Magill, M. J. and Constantinides, G. M. (1976), ‘Portfolio selection with transactions costs’, *Journal of Economic Theory* **13**(2), 245–263.
- Markowitz, H. (1952), ‘Portfolio selection’, *The Journal of Finance* **7**(1), 77–91.
- Mei, X., DeMiguel, V. and Nogales, F. J. (2016), ‘Multiperiod portfolio optimization with multiple risky assets and general transaction costs’, *Journal of Banking & Finance* **69**, 108–120.
- Merton, R. C. (1969), ‘Lifetime portfolio selection under uncertainty: The continuous-time case’, *The Review of Economics and Statistics* pp. 247–257.

- Merton, R. C. (1971), ‘Optimum consumption and portfolio rules in a continuous-time model’, *Journal of Economic Theory* **3**(4), 373 – 413.
- Merton, R. C. (1973), ‘An intertemporal capital asset pricing model’, *Econometrica: Journal of the Econometric Society* pp. 867–887.
- Modigliani, F. and Miller, M. H. (1958), ‘The cost of capital, corporation finance and the theory of investment’, *The American Economic Review* **48**, 261–297.
- Muthuraman, K. and Kumar, S. (2006), ‘Multidimensional portfolio optimization with proportional transaction costs’, *Mathematical Finance: An International Journal of Mathematics, Statistics and Financial Economics* **16**(2), 301–335.
- Nielsen, S. S. and Poulsen, R. (2004), ‘A two-factor, stochastic programming model of danish mortgage-backed securities’, *Journal of Economic Dynamics and Control* **28**(7), 1267–1289.
- Olivares-Nadal, A. V. and DeMiguel, V. (2018), ‘A robust perspective on transaction costs in portfolio optimization’, *Operations Research* .
- Parpas, P., Ustun, B., Webster, M. and Tran, Q. K. (2015), ‘Importance sampling in stochastic programming: A markov chain monte carlo approach’, *INFORMS Journal on Computing* **27**(2), 358–377.
- Pennanen, T. and King, A. J. (2004), *Arbitrage pricing of American contingent claims in incomplete markets - a convex optimization approach*, Humboldt-Universität zu Berlin, Mathematisch-Naturwissenschaftliche Fakultät II, Institut für Mathematik.
- Pflug, G. (2001), ‘Scenario tree generation for multiperiod financial optimization by optimal discretization’, *Mathematical Programming* **89**(2), 251–271.
- Pflug, G. C. and Broussev, N. (2009), ‘Electricity swing options: Behavioral models and pricing’, *European Journal of Operational Research* **197**(3), 1041–1050.
- Powell, W. B. (2011), *Approximate dynamic programming: Solving the curses of dimensionality*, Wiley, Hoboken, New Jersey.
- Powell, W. B. (2014), *Clearing the Jungle of Stochastic Optimization*, INFORMS, chapter 4, pp. 109–137.
- Powell, W. B. (2016), *A Unified Framework for Optimization Under Uncertainty*, INFORMS, chapter 3, pp. 45–83.
- Powell, W. B. (2018), A unified framework for stochastic optimization. Preprint submitted to European J. Operation Research.
- Puterman, M. L. (2005), *Markov decision processes: Discrete stochastic dynamic programming*, Wiley, Hoboken, New Jersey.

- Rasmussen, K. M. and Clausen, J. (2007), ‘Mortgage loan portfolio optimization using multi-stage stochastic programming’, *Journal of Economic Dynamics and Control* **31**(3), 742–766.
- Rogers, L. C. (2002), ‘Monte carlo valuation of american options’, *Mathematical Finance* **12**(3), 271–286.
- Shapiro, A., Dentcheva, D. and Ruszczyński, A. (2014), *Lectures on Stochastic Programming: Modeling and Theory*, 2 edn, Society for Industrial and Applied Mathematics.
- Steinbach, M. C. (2003), ‘Tree-sparse convex programs’, *Mathematical Methods of Operations Research* **56**(3), 347–376.
- Topaloglou, N., Vladimirov, H. and Zenios, S. A. (2008a), ‘A dynamic stochastic programming model for international portfolio management’, *European Journal of Operational Research* **185**(3), 1501–1524.
- Topaloglou, N., Vladimirov, H. and Zenios, S. A. (2008b), ‘Pricing options on scenario trees’, *Journal of Banking & Finance* **32**(2), 283–298.
- Tsitsiklis, J. N. and Van Roy, B. (2001), ‘Regression methods for pricing complex american-style options’, *IEEE Transactions on Neural Networks* **12**(4), 694–703.
- van Binsbergen, J. H. and Brandt, M. W. (2007), ‘Solving dynamic portfolio choice problems by recursing on optimized portfolio weights or on the value function?’, *Computational Economics* **29**(3-4), 355–367.
- Van Slyke, R. M. and Wets, R. (1969), ‘L-shaped linear programs with applications to optimal control and stochastic programming’, *SIAM Journal on Applied Mathematics* **17**(4), 638–663.
- Volosov, K., Mitra, G., Spagnolo, F. and Lucas, C. (2005), ‘Treasury management model with foreign exchange exposure’, *Computational Optimization and Applications* **32**, 179–207.

Papers

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